

Sina Ronaghi

Metrologist | Multidisciplinary R&D Expert | Leader
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LinkedIn | ResearchGate | GitHub | Online Portfolio

PROFILE SUMMARY

Multidisciplinary Electrical Engineer (Ph.D.) specialized in **Metrology**, **Data Science**, **Embedded Systems**, and **Signal Processing**. Proven track record within the high-tech industrial and R&D settings with a particular focus on entrepreneurship and project management. **Experienced leader** in startups and EU Flagship Programs with a hands-on and engaging attitude. Driven by curiosity and passionate about adopting **emerging AI technologies** for process optimization, task coordination, and improving team efficiency.

CORE COMPETENCIES

Signal Processing	Waveform Analysis, Harmonic Characterization and THD Analysis, Digital & Statistical Signal Processing, Sensor Fusion.
Engineering Tools	Python , MATLAB (SIMULINK) & R (Predictive Maintenance, Machine Learning, Data Visualization & Signal Processing), Databases (SQL, MongoDB, InfluxDB), Circuit Design & Analysis (Altium, LTSpice), Instrumentation (NI LabVIEW/TestStand), Embedded (C/C++, RTOS, Bare-Metal, Arduino/ARM)
Metrology Framework	JCGM/GUM, CENELEC, WELMEC, EN, IEC, MID
Regulatory Exposure	EN 50160 (Voltage Characteristics), IEC 62052/53/56 (Smart Metering), IEC 61000 (EMC), ISO/IEC 17025 (Testing, Inspection) & Certification, ISO 9001 (QMS)
AI & Automation	Vertex AI, n8n, Node-RED, Ollama, Notebook LM, AWS, AZURE & GCP
Soft Skills	Certified LEAN Manager (Kaizen, 5S), Agile & DevOps (Trello, Jira & GitLab), Proactive, Public speaking, Cross-disciplinary stakeholder engagement

PROFESSIONAL EXPERIENCE

Post-Doc Researcher / Project Manager	Nov 2025 – Present
Politecnico di Milano	Milan, Italy
- Optimized R&D testing frameworks for VT/CT characterization for LV/MV voltage characteristics (EN 50160) for Smart Grid Optimization in MATLAB/Python, increasing testing throughput by 70%. - Directed ISO-compliant lab operations (IEC 17025 & ISO 9001), integrating LEAN management (Kaizen & 5S) to enhance workplace safety, and R&D output. - Managed R&D proposal drafting for EU Flagship Programs (Erasmus+, Horizon) using Agile methodologies (Trello, Jira, GitLab) to synchronize milestones across 15+ EMEA partners (e.g., Infineon, Imec, Siemens). - Integrated AI-driven tools (Vertex AI, LLMs) into HW/SW workflows, automating R&D processes to accelerate development cycles.	
R&D Associate Manager - Sustainable Mobility	Nov 2024 – Oct 2025
Politecnico di Milano	Milan, Italy
Recruited to lead a research project on electric vehicles (EV), while concurrently finalizing my Ph.D. dissertation.	
- Led the design of a sensor fusion system for EV navigation using MEMS-Based IMUs, integrating Bluetooth Low Energy on ARM-based embedded HW in C (nRF) - Automated test sessions by using NI LabVIEW/TestStand, reducing 50% of the operational testing time for EV navigation tests compared to the legacy solution while ensuring measurement repeatability. - Directed data-driven EV navigation predictive maintenance, employing statistical learning and anomaly detection to ensure system reliability.	
Ph.D. Researcher - Green Energy Integration & Smart Metering	Nov 2021 – May 2025
Politecnico di Milano	Milan, Italy
Innovated novel predictive models for Smart Metering compliance and anomaly detection (IEC 62052/53/56)	

via best practice machine learning, and advanced signal processing with > 90% accuracy.

- Deployed a scalable IIoT pipeline to monitor 2G Smart Meters via MQTT and cloud microservices (Grafana, InfluxDB) on AWS EC2 servers, ensuring 95%+ data integrity across 3+ years of continuous operation.
- Participated in research for Stress Monitoring via Wearable BioSignal Sensors (PPG & EEG). Applied waveform and THD analysis, and defined Novel Signal Quality Assessment (SQA) indicator to enhance the reliability of sensor data in Med-Tech.
- Formulated standardized metrological decision frameworks, contributing to international JCGM guidance for conformity assessment and risk-based decision frameworks.

Technical Consultant

Sep 2018 – Sep 2019

Freelance contractor [Startup ecosystems]

Ahvaz, Iran

- Deployed IIoT monitoring systems (M2M), optimizing ARM-based embedded hardware for EMC Compliance (IEEE 61000) in noisy environments using Embedded C, Altium Designer & LTspice.
- Designed sensor fusion dashboards, leveraging software-driven control systems for real-time monitoring (RTOS) and data acquisition in the Agri-Tech sector via Arduino, NI LabVIEW, and Node-RED.
- Developed scalable software architectures (Node.js/MongoDB), implementing secure RESTful APIs to manage high-scale data and user authentication for online multiplayer games.

Co-Founder & CEO

Dec 2017 – Aug 2018

Atiyeh Hoshyar LLC. [Startup]

Ahvaz, Iran

- Co-founded an IoT startup and secured 60% of total funding through targeted pitches and incubator programs, accelerating early product development and market validation.
- Scaled engineering capacity by 40% in the first year by recruiting, onboarding, and leading a 5-person R&D team specializing in electronic design and embedded hardware/firmware.
- Filed a patent in Home Automation and Remote Monitoring, integrating secure M2M communication into ARM-based embedded HW/FW architectures.

EDUCATION

Ph.D. in Electrical Engineering (with honors)

Nov 2021 – May 2025

Politecnico di Milano

Milan, Italy

Thesis (public): Data-driven approaches for anomaly detection in energy metering.

M.Sc. in Electrical Engineering

Sep 2019 – Oct 2021

Politecnico di Milano

Milan, Italy

Thesis (public): Real-time motion capturing using inertial measurement units and applied embedded systems development

B.Sc. in Electrical Engineering

Sep 2013 – Jul 2018

Shahid Chamran University

Ahvaz, Iran

Thesis: Embedded Systems Design for Test Automation in AgriTech.

ADDITIONAL INFORMATION

Certificates	LEAN Management Fundamentals, Probability and Data with R, Python Data Structures, Data Scientist’s ToolBox (Coursera); NI LabVIEW Object-Oriented Programming (Emerson); Signal and Power Integrity for High-Speed PCBs (Polimi); EU Driving License (B).
Awards	IEEE AMPS 2023 - Best Presentation Award; MIUR Scholarship for PhD Studies.
Committees	Technical Program Committee (TPC) for IEEE MetroXRAINE (since 2024).
Memberships	GMEE, IEEE Instrumentation & Measurement society (since 2021).
Languages	English (C2); Italian (B1).
Visa Sponsorship	Not required